Model diagnostics and interpretation

Example 1

We want to fit the data with poisson GLM.

Recall that

$$\beta_0 + \beta_1 x_i = \eta_i = g(\mu_i) = \log(\mu_i)$$

```
x<-c(1,2)
y<-c(0,1)
model1<-glm(y~ x-1,family=poisson(link="log"))
summary(model1)
```

```
##
## Call:
## glm(formula = y ~ x - 1, family = poisson(link = "log"))
##
## Coefficients:
##   Estimate Std. Error z value Pr(>|z|)
## x  -0.2475     0.5573  -0.444    0.657
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 2.0000  on 2  degrees of freedom
## Residual deviance: 1.7706  on 1  degrees of freedom
## AIC: 5.7706
##
## Number of Fisher Scoring iterations: 5
```

```
coefficient<-summary(model1)$coefficients
temp<-exp(coefficient[1,1]*x) #Fitted value of mu
temp
```

```
## [1] 0.7807764 0.6096118
```

```
summary(model1)$deviance #Deviance residual
```

```
## [1] 1.770642
```

```
residuals(model1,type="pearson")
```

```
##          1          2
## -0.8836155  0.5000000
```

```
rstandard(model1,type="pearson")
```

```
##         1         2
## -1.015272  1.015272
```

```
lmodel<-logLik(model1)
lnull<-logLik(update(model1,.~1))
```

```
rsquared<-1-(lmodel/lnull)
rsquared
```

```
## 'log Lik.' -0.113501 (df=1)
```

The fitted means are similar (($\hat{\mu}_1 \approx 0.78$), ($\hat{\mu}_2 \approx 0.61$)), the standardised Pearson residuals ($\approx \pm 1.02$) are not especially large, and the deviance-based pseudo-R^2 is low (≈ 0.114), indicating only a small improvement over the null model, though conclusions are limited by the small sample size.

Example 2

```
library(faraway)
data(gala)
str(gala)
```

```
## 'data.frame':    30 obs. of  7 variables:
##  $ Species  : num  58 31 3 25 2 18 24 10 8 2 ...
##  $ Endemics : num  23 21 3 9 1 11 0 7 4 2 ...
##  $ Area     : num  25.09 1.24 0.21 0.1 0.05 ...
##  $ Elevation: num  346 109 114 46 77 119 93 168 71 112 ...
##  $ Nearest  : num  0.6 0.6 2.8 1.9 1.9 8 6 34.1 0.4 2.6 ...
##  $ Scruz    : num  0.6 26.3 58.7 47.4 1.9 ...
##  $ Adjacent : num  1.84 572.33 0.78 0.18 903.82 ...
```

```
temp<-gala[,-2]
str(temp)
```

```
## 'data.frame':    30 obs. of  6 variables:
##  $ Species  : num  58 31 3 25 2 18 24 10 8 2 ...
##  $ Area     : num  25.09 1.24 0.21 0.1 0.05 ...
##  $ Elevation: num  346 109 114 46 77 119 93 168 71 112 ...
##  $ Nearest  : num  0.6 0.6 2.8 1.9 1.9 8 6 34.1 0.4 2.6 ...
##  $ Scruz    : num  0.6 26.3 58.7 47.4 1.9 ...
##  $ Adjacent : num  1.84 572.33 0.78 0.18 903.82 ...
```

```
model1<-glm(Species~.,family=poisson(link="log"),data=gala)
model2<-summary(model1)
names(model2)
```

```
##  [1] "call"           "terms"          "family"         "deviance"
##  [5] "aic"            "contrasts"      "df.residual"    "null.deviance"
##  [9] "df.null"        "iter"           "deviance.resid" "coefficients"
## [13] "aliased"        "dispersion"     "df"             "cov.unscaled"
## [17] "cov.scaled"
```

```
model2$deviance
```

```
## [1] 313.3558
```

```
set1<-c("Elevation","Nearest","Scruz","Adjacent","1")
for (i in set1){
    nullmodel<-update(model1,as.formula(paste(". ~ . -", i)))
    print(anova(nullmodel,model1,test="Chisq"))
}
```

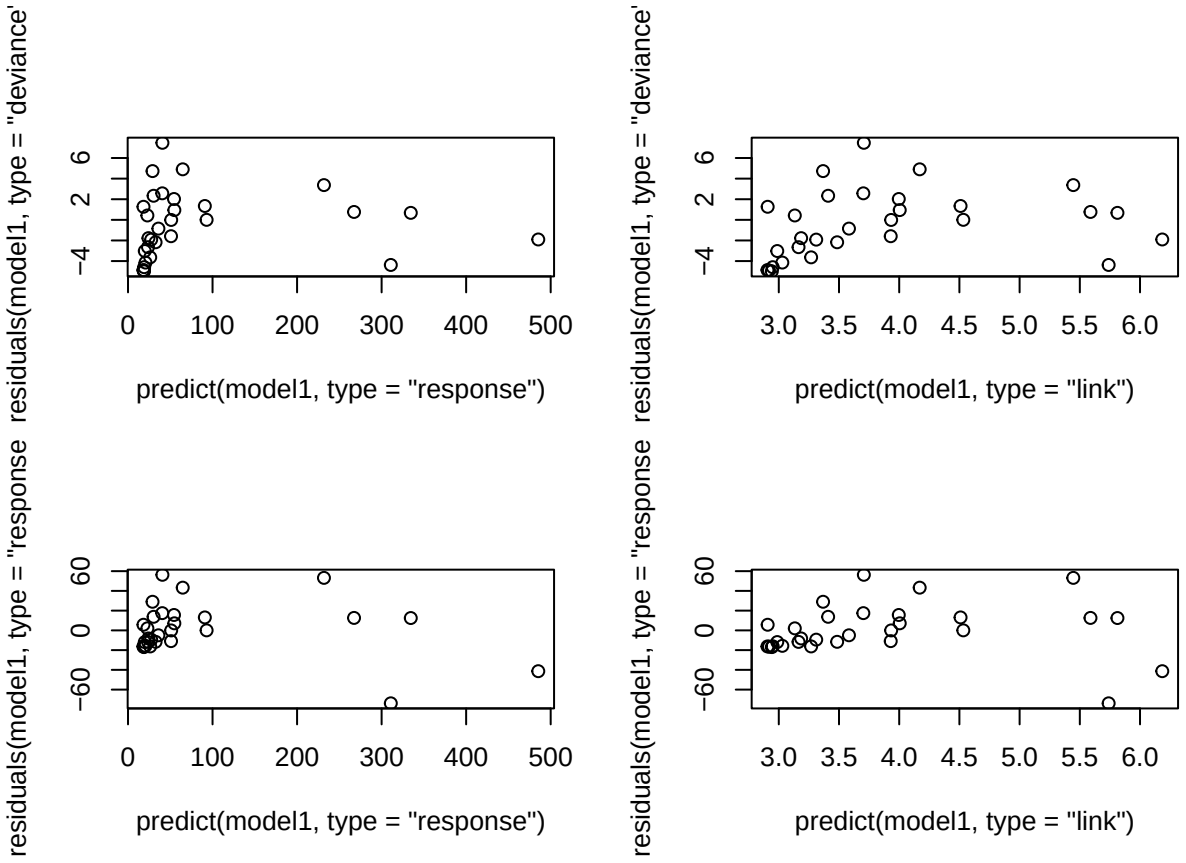
```
## Analysis of Deviance Table
##
```

```
## Model 1: Species ~ Endemics + Area + Nearest + Scruz + Adjacent
## Model 2: Species ~ Endemics + Area + Elevation + Nearest + Scruz + Adjacent
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1        24     315.20
## 2        23     313.36  1   1.8481    0.174
## Analysis of Deviance Table
##
## Model 1: Species ~ Endemics + Area + Elevation + Scruz + Adjacent
## Model 2: Species ~ Endemics + Area + Elevation + Nearest + Scruz + Adjacent
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1        24     353.21
## 2        23     313.36  1    39.85 2.743e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## Analysis of Deviance Table
##
## Model 1: Species ~ Endemics + Area + Elevation + Nearest + Adjacent
## Model 2: Species ~ Endemics + Area + Elevation + Nearest + Scruz + Adjacent
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1        24     314.77
## 2        23     313.36  1   1.4182   0.2337
## Analysis of Deviance Table
##
## Model 1: Species ~ Endemics + Area + Elevation + Nearest + Scruz
## Model 2: Species ~ Endemics + Area + Elevation + Nearest + Scruz + Adjacent
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1        24     314.25
## 2        23     313.36  1  0.89524   0.3441
## Analysis of Deviance Table
##
## Model 1: Species ~ Endemics + Area + Elevation + Nearest + Scruz + Adjacent -
##     1
## Model 2: Species ~ Endemics + Area + Elevation + Nearest + Scruz + Adjacent
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1        24    1651.46
## 2        23     313.36  1   1338.1 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
fullmodel<-logLik(model1)
nullmodel<-logLik(update(model1,.~1))
1-fullmodel/nullmodel
```

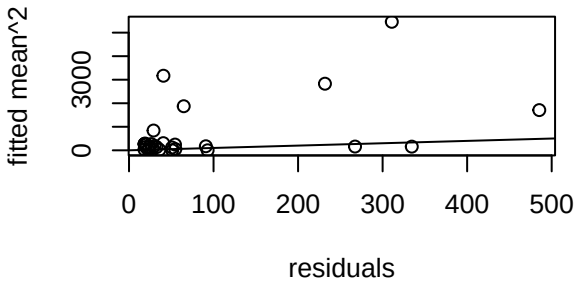
```
## 'log Lik.' 0.8708487 (df=7)
```

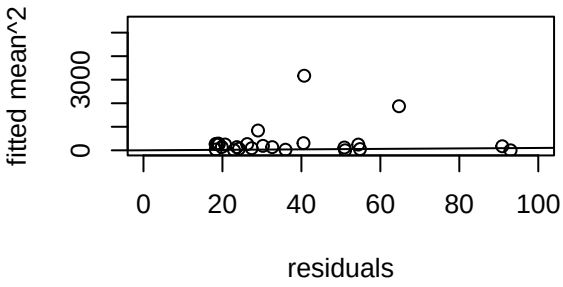
```
par(mfrow=c(2,2))
plot(predict(model1,type="response"),residuals(model1,type="deviance"))
plot(predict(model1,type="link"),residuals(model1,type="deviance"))
plot(predict(model1,type="response"),residuals(model1,type="response"))
plot(predict(model1,type="link"),residuals(model1,type="response"))
```

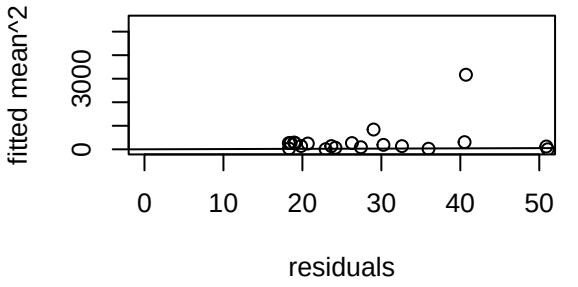


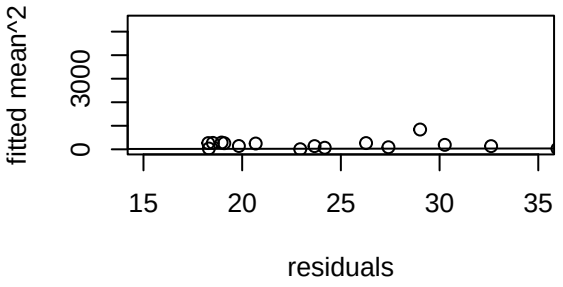
Within each row, the two plots show the same residuals against fitted values on different scales: fitted mean and linear predictor. The first row shows deviance residuals, and the second row shows raw response residuals. The residuals mostly cluster around zero, and there is no obvious smooth trend. However, there are few observations at higher fitted values, and these plots alone are not enough to confirm that the model is adequate. Therefore, we should also check standardised residuals, dispersion and influential observations.

```
library(faraway)
data(gala)
model1<-glm(Species~.,family=poisson(link="log"),data=gala)
mu<-fitted(model1,type="response")
res<-residuals(model1,type="response")
par(mfrow=c(2,2))
plot(mu,res^2,xlab="residuals",ylab="fitted mean^2")
abline(a=0,b=1)
plot(mu,res^2,xlab="residuals",ylab="fitted mean^2",xlim=c(0,100))
abline(a=0,b=1)
plot(mu,res^2,xlab="residuals",ylab="fitted mean^2",xlim=c(0,50))
abline(a=0,b=1)
plot(mu,res^2,xlab="residuals",ylab="fitted mean^2",xlim=c(15,35))
abline(a=0,b=1)
```









The squared-residual-versus-fitted-mean plot can be compared with the Poisson reference line $y = x$. Several observations lie well above the line, suggesting larger-than-expected squared residuals for those observations. This may indicate unusual observations or extra variability, but the plot alone is not enough to conclude that the Poisson model fits or fails to fit.

Recall that for a GLM, the adjusted dependent variable is defined as

$$z_i = \hat{\eta}_i + (y_i - \hat{\mu}_i) \left.\frac{\partial \eta_i}{\partial \mu_i}\right|_{\mu_i = \hat{\mu}_i}$$

For a Poisson GLM with a log link,

$$\eta_i = \log(\mu_i)$$

Therefore,

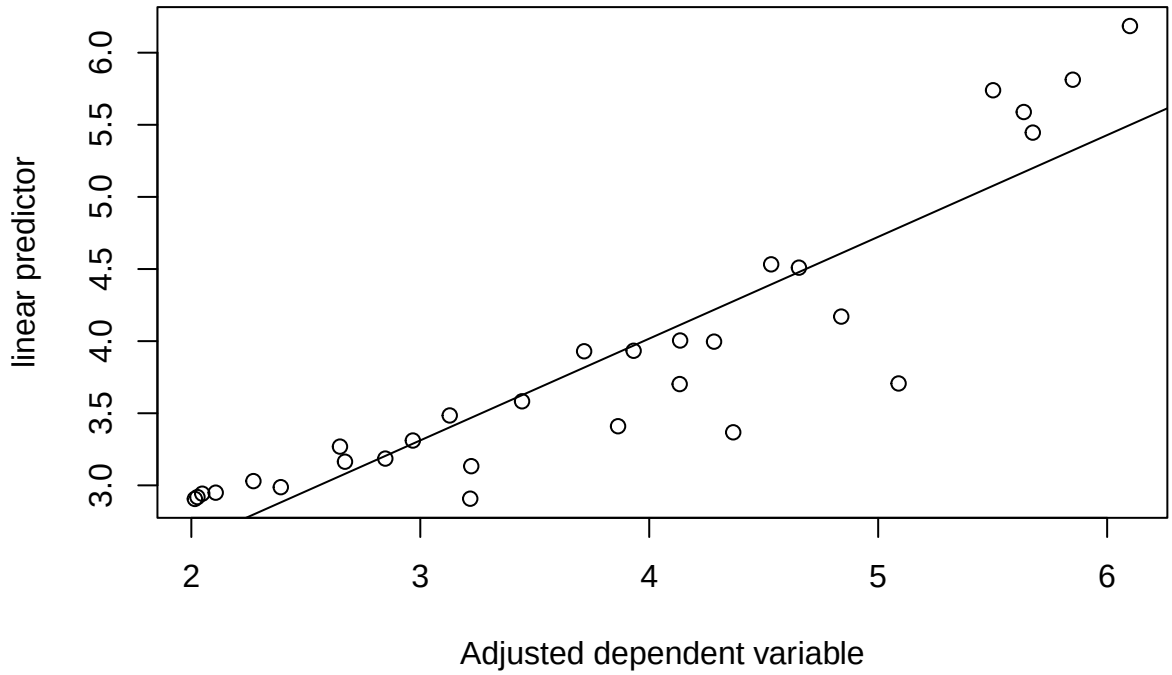
$$\boxed{z_i = \log(\hat{\mu}_i) + \frac{y_i - \hat{\mu}_i}{\hat{\mu}_i}}.$$

```
library(faraway)
data(gala)
model1<-glm(Species~.,family=poisson(link="log"),data=gala)
mu<-fitted(model1,type="response")
y<-gala$Species
z<-log(mu)+(y-mu)/mu
plot(z,log(mu),xlab="Adjusted dependent variable",ylab="linear predictor")
temp<-lm(log(mu)~z)
temp
```

```
##
```

```
## Call:
## lm(formula = log(mu) ~ z)
##
## Coefficients:
## (Intercept)            z
##      1.1949       0.7056
```

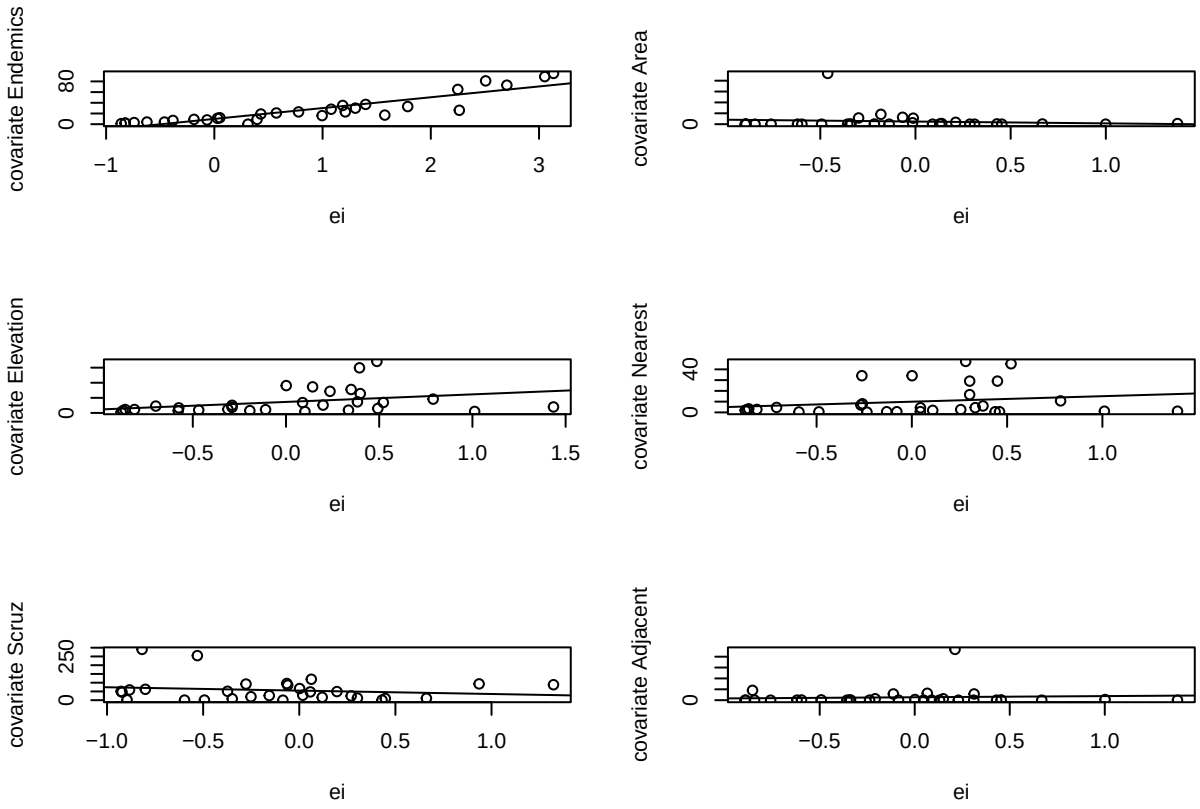
```
x<-seq(1,8,by=0.001)
lines(x,temp$coefficients[1]+temp$coefficients[2]*x)
```



The adjusted dependent variable shows an approximately linear relationship with the fitted linear predictor, although there is noticeable scatter around the trend. There is no obvious strong curvature, which provides some evidence that the log link is reasonable for this Poisson GLM, but it does not prove the link is correct.

```
library(faraway)
data(gala)
model1<-glm(Species~.,family=poisson(link="log"),data=gala)
mu<-fitted(model1,type="response")
y<-gala$Species
z<-log(mu)+(y-mu)/mu
x<-model.matrix(model1)
eta<-predict(model1,type="link")
par(mfrow=c(3,2))
set1<-c("Endemics","Area","Elevation","Nearest","Scruz","Adjacent")
for (i in set1){
temp4<-z-eta+coef(model1)[i]*x[,i]
    plot(temp4,x[,i],ylab=paste("covariate",i),xlab="ei")
```

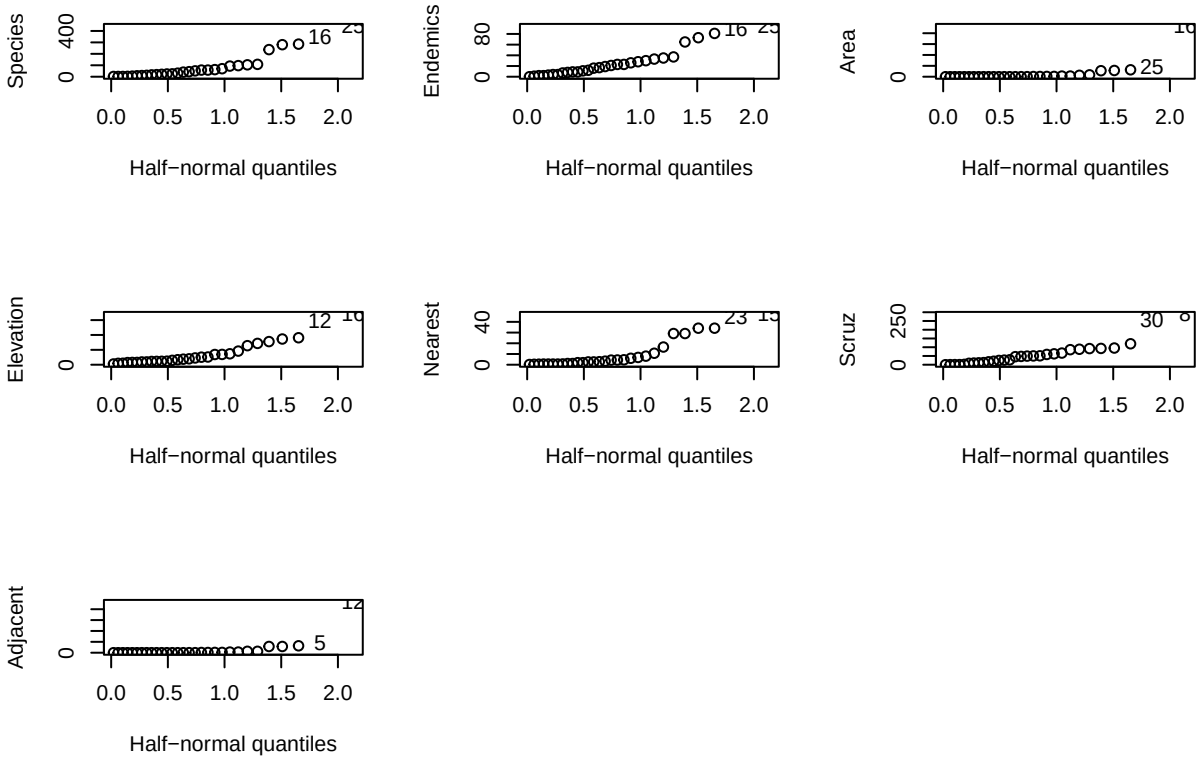
```
    temp<-lm(x[,i]~temp4)
    x2<-seq(-10,10,by=0.001)
    lines(x2,temp$coefficients[1]+temp$coefficients[2]*x2)
}
```



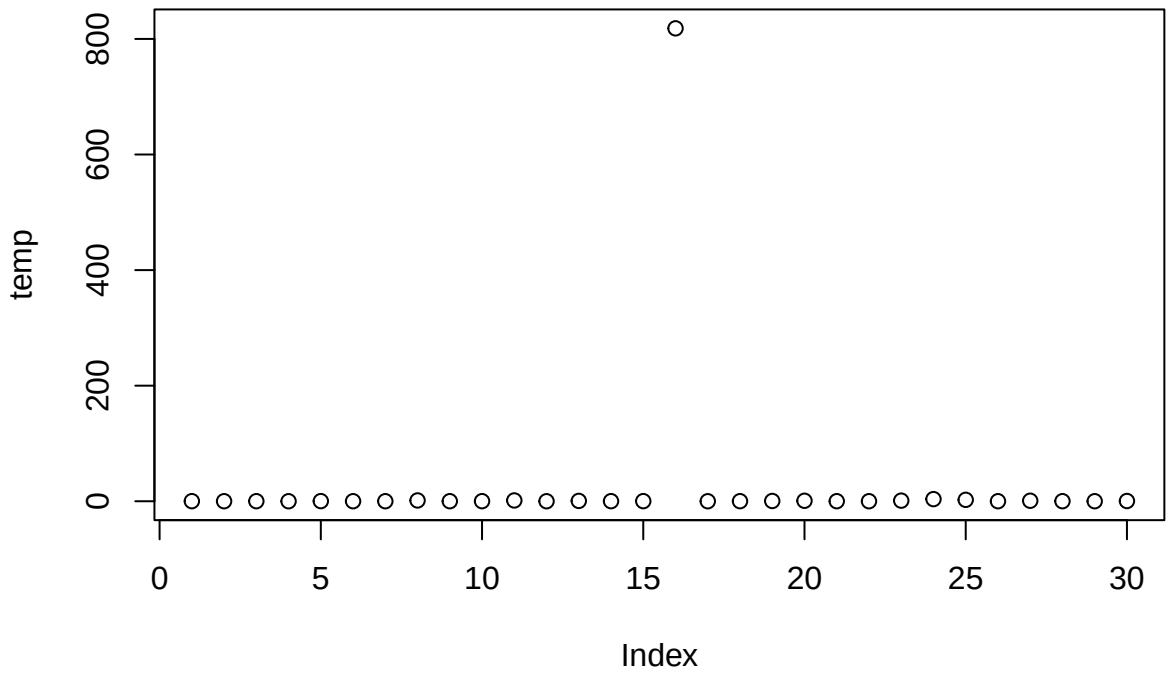
Under the correct link function, the plot should be approximately linear. For Endemics, Area, and Adjacent, the observations lie very close to the straight reference line. The other variables also appear fairly close to linearity, although with a bit more scatter. Overall, there is no strong evidence of curvature, so a Poisson GLM with a log link seems reasonable for this data.

```
library(faraway)
data(gala)
Xmatrix<-gala[,2:dim(gala)[2]]
Ymatrix<-gala[,1,drop=FALSE]
colnames(Ymatrix)<-"Species"
par(mfrow=c(3,3))
temp<-cbind(Ymatrix,Xmatrix)
for (i in 1:dim(temp)[2]){
    halfnorm(temp[,i],ylab=colnames(temp[i]))

}
temp<-cooks.distance(model1)
par(mfrow=c(1,1))
```


Species
0
400
16
0.0
0.5
1.0
1.5
2.0
Half−normal quantiles
Endemics
0
80
16
0.0
0.5
1.0
1.5
2.0
Half−normal quantiles
Area
0
25
0.0
0.5
1.0
1.5
2.0
Half−normal quantiles
Elevation
0
12
0.0
0.5
1.0
1.5
2.0
Half−normal quantiles
Nearest
0
40
23
0.0
0.5
1.0
1.5
2.0
Half−normal quantiles
Scruz
0
250
30
0.0
0.5
1.0
1.5
2.0
Half−normal quantiles
Adjacent
0
5
0.0
0.5
1.0
1.5
2.0
Half−normal quantiles

```
plot(temp)
```




```
temp
```

```
##       Baltra    Bartolome     Caldwell     Champion      Coamano Daphne.Major
## 1.435521e-01 1.560872e-02 9.532395e-02 1.555453e-03 2.107271e-01 3.586857e-02
## Daphne.Minor       Darwin         Eden      Enderby     Espanola   Fernandina
## 1.682451e-02 1.376062e+00 7.268074e-02 1.047062e-01 1.387991e+00 7.820974e-04
##     Gardner1     Gardner2     Genovesa      Isabela     Marchena       Onslow
## 5.621409e-01 1.083858e-01 1.607682e-01 8.183238e+02 1.024188e-06 1.034444e-01
##        Pinta       Pinzon   Las.Plazas       Rabida SanCristobal  SanSalvador
## 4.427323e-01 7.937171e-01 6.270759e-02 4.811773e-02 1.068509e+00 3.696020e+00
##    SantaCruz      SantaFe   SantaMaria      Seymour      Tortuga         Wolf
## 2.434974e+00 1.408071e-02 8.703878e-01 6.581649e-02 2.064002e-02 4.107203e-01
```

Since we are interested in the magnitude of the effects of the observations, we use a half-normal plot. From the plot, we found that Species, Nearest, Endemics, and Scruz have relatively large effects, while the other variables are smaller. The effects increase as the quantiles increase, and the increase is more pronounced for the larger quantiles. We also found that for the larger quantiles, the points are far from the reference line, which suggests that these observations have a greater effect on the GLM.

From the Cook's distance plot, it is clear that only one observation, Isabela, has a very large Cook's distance, indicating that it has a substantial influence on the model. Fernandina, on the other hand, has a very small Cook's distance, indicating very little influence on the model. These results suggest that most observations contribute fairly evenly to the model, while only a few observations have unusually large or small influence.

Based on the diagnostic plots, the Poisson GLM with log link appears to be a reasonable model for the data. The residual plots do not show strong systematic patterns, the link function seems appropriate, and most observations contribute fairly evenly to the model. However, there is one influential observation, Isabela, so the model should be interpreted with some caution.